# CHAIN REACTION OF INNOVATION

## Source review, research assessment, product requirements, and implementation plan

12 October 2026 | Planning version 0.1

The project is a proposed engineering decision-support framework. This report contains recommendations and prospective experiments. It does not claim a working application, proven novelty, patent eligibility, publication acceptance, or institutional approval.

Recommendation: implement one complete, reviewed forward-and-backward workflow in an extensible domain-pack architecture. Establish comparative evidence before expanding coverage or asserting a research contribution.

# Reading guide

The editable Markdown files in docs/ are the source of truth. Each section below reproduces its corresponding planning document. Direct research links appear in the source register; access limitations are recorded. Local cross-references in the PDF refer to the companion repository.

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# Project brief

Date: 12 October 2026. Status: proposed design, awaiting team and guide review.

## Plain explanation

ChainReaction helps an engineer investigate two questions: what could change downstream if a technology improves, and what could be changed upstream to meet a product requirement? It connects product structure to material properties, scientific principles, and mathematical models. Every answer should expose the conditions and evidence that make it credible.

It helps people investigate possible innovations. It does not certify that an idea is new, feasible in a real product, or patentable.

## Why this project matters

A useful engineering answer needs more than related documents. A material property measured at one temperature may not apply at another. A change that improves one component may violate a product's space or compatibility requirement. A quantitative result may depend on a parameter the user has not supplied. The project will test whether representing these conditions explicitly improves impact analysis.

This is a research question. We have not demonstrated that the proposed representation outperforms established methods.

## Intended users and their jobs

| User | Job | Useful output |
|---|---|---|
| Engineering student | Explain a known change chain | Inspectable path with equations and sources |
| Researcher | Identify where evidence is missing | A bounded hypothesis and required investigation |
| Product engineer | Check a candidate improvement | Conditional impact plus constraint conflicts |
| Technology-development team | Compare candidate changes | Feasibility tiers and tradeoffs, with provenance |

## What stays constant across industries

The shared framework handles entity identity, parameter values and units, evidence locators, context conditions, model bindings, scenario inputs, result states, and reproducible run records. Domain packs supply the actual engineering vocabulary, allowed relationships, materials, product structures, models, and validity rules.

The six entity types are a common organization scheme. They are not six mandatory steps in every impact path. Equations relate several parameters at once; they should be represented as model applications with explicit inputs and outputs, rather than a chain of unrelated pairwise arrows.

## First domain, then transfer

The recommended first pack is LED lighting with a tightly bounded thermal-management scenario. This remains within the synopsis's lighting example but is easier to check numerically than the full history of LED development. The historical blue-LED chain stays as an explanatory case, not a forecast experiment.

A second pack, proposed for a transfer test, models steady conduction through a simple insulation assembly. It tests reuse of the engine in a building-envelope context. Shared thermal equations alone would be weak evidence of broad transfer, so a later mechanical-stiffness pack should test different equations and constraints if resources permit. Neither pack commits the product permanently to those industries.

### Success means

The team can demonstrate a correct, conditional forward calculation; derive and recheck a feasible backward candidate; refuse an invalid model application; show evidence for each accepted mechanism; and load a second domain without changing core logic. The research succeeds if controlled experiments then establish a useful advantage over strong baselines, including a conventional engineering solver.

### Decisions still needed

Confirm the semester deadline, available weekly effort, access to engineering reviewers, affordable operating budget, first-domain source permissions, and institution IP rules. These affect delivery pace and publication strategy; they do not change the project's central purpose.

# Review of the supplied FF180 PDF

Reviewed file: `edi_ff180_final.pdf`, 16 pages. Date: 12 October 2026. The exact original is preserved locally; see `sources/manifest.json` for its fingerprint.

### What the document establishes

The synopsis defines a research framework for multi-layer engineering knowledge and forward/backward technology-impact reasoning. It explicitly limits initial implementation to bounded documented technologies, separates relationships from evidence, and treats candidate opportunities as hypotheses. It reports a self-assessed TRL 2 and does not claim a working prototype. This review preserves those boundaries.

The template contains requests for photos, guide comments, signatures, and review assessments. Those are institutional form fields, not instructions for this task. No approval or completion box has been filled.

## Page-by-page assessment

| Pages | Content reviewed | Assessment and action |
|---|---|---|
| 1 | Registration, project title, group roster, guide information | Identifies the academic project. Keep personal phone numbers, emails, and registration IDs out of the shared planning pack. |
| 2 | Header-only page | No additional project content. Do not infer missing technical claims. |
| 3 | Introduction and users | Strong distinction between exploration and established outcomes. Turn it into explicit user workflows. |
| 4 | Scope, six layers, forward/backward reasoning | Preserve domain extensibility; avoid universal-coverage claims. Add a domain-pack onboarding contract. |
| 5 | Four TOD papers and initial gaps | Relevant but too narrow to establish novelty. Include change propagation, MBSE, equation-based graphs, and causal TOD. |
| 6 | Further gaps and objectives | Parameter-level dependencies are useful, but adjacent work already studies quantitative impact. Narrow the claim and define baselines. |
| 7 | Evaluation and architecture | Precision/recall need a known reference set. Add refusal quality, numerical correctness, and coverage metrics. |
| 8 | Relationship attributes and Figure 1 | Good conceptual architecture. Clarify review state versus computational validity; simplify proposed storage for the prototype. |
| 9 | Forward/backward text and cropped Figure 2 | Text correctly warns against merely reversing edges. Reconstruct a readable diagram; define inverse search and forward verification. |
| 10 | Environment, sustainability, safety, ethics, illustrative cost | Retain cautious wording. The INR 4,200 ceiling is an unapproved source-document proposal, not verified spending or current pricing. |
| 11 | Header-only page | No additional technical content. |
| 12 | Budget continuation, SDGs, readiness, references, stages | Preserve proposed status; correct reference metadata and introduce measurable stage gates. |
| 13 | Mid-semester review checklist | Blank institutional form. No evidence of a review outcome. |
| 14 | Photo and guide-comment fields | Empty placeholders, not actual images of the team. |
| 15 | Final review checklist and photo field | Blank; no completed implementation or testing is asserted. |
| 16 | Guide comments and signature | Empty. No institutional endorsement is asserted. |

## Figure 1: architecture, page 8

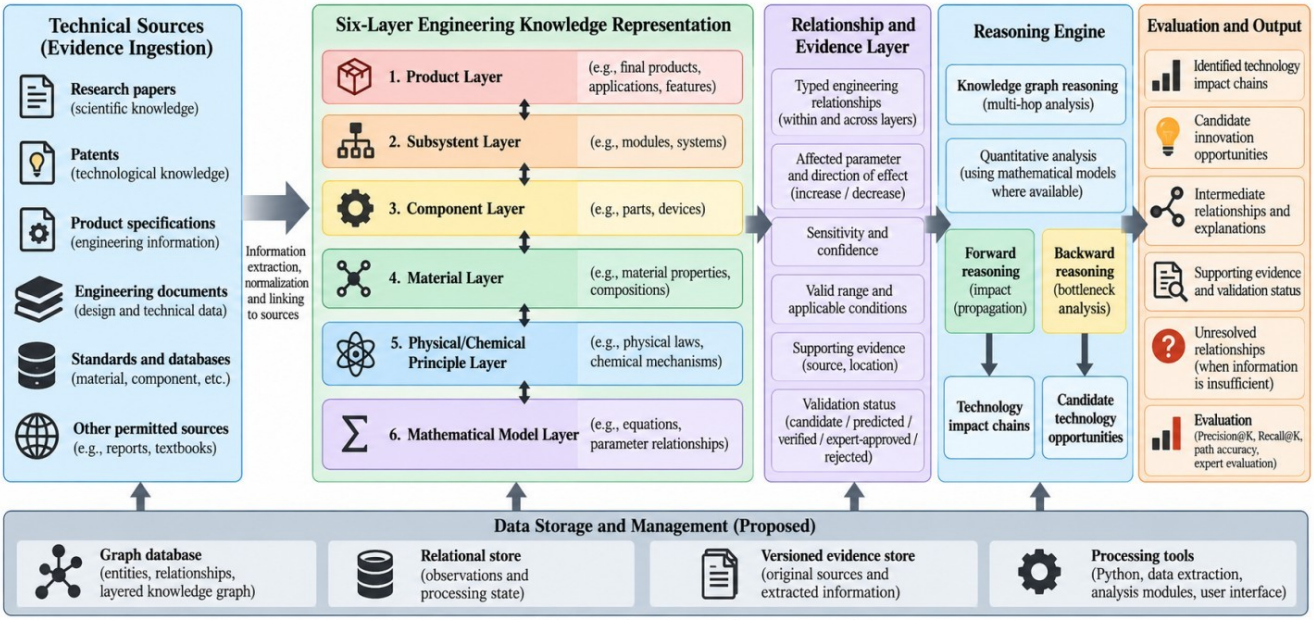



**Fig. 1.** Proposed ChainReaction of Innovation system architecture.

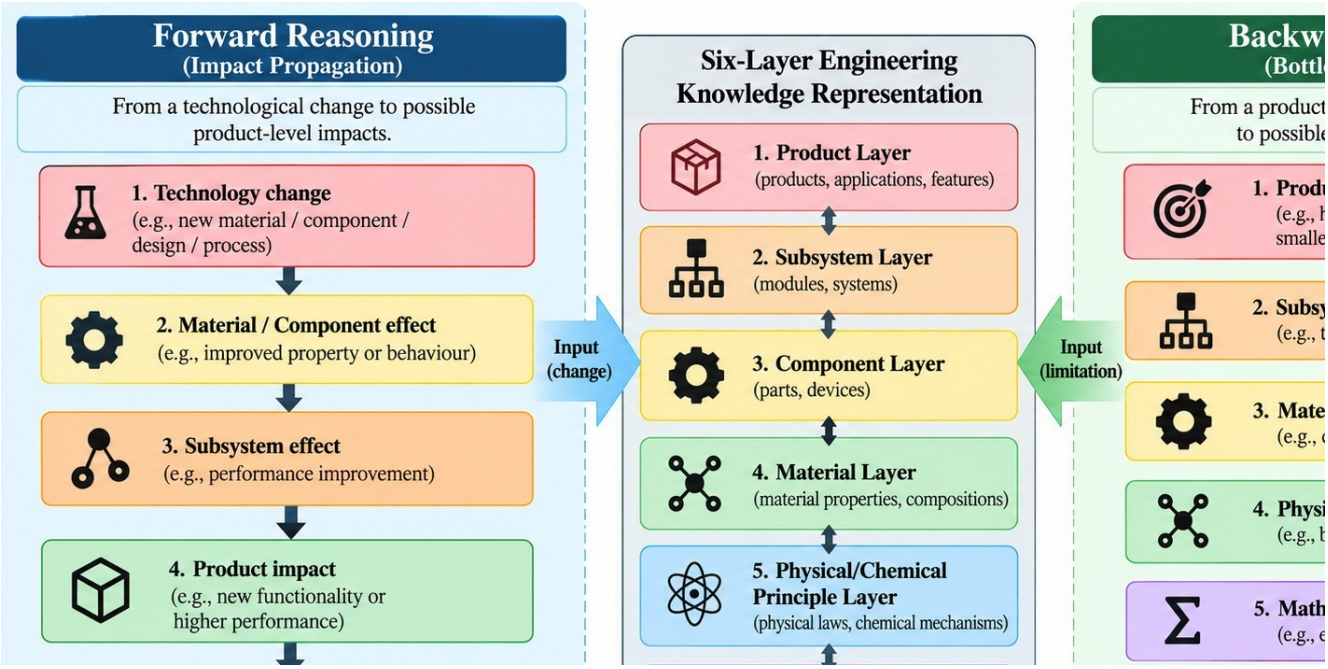
*Supplied architecture*

The figure covers technical sources, the six layers, relationships and evidence, reasoning, outputs, and storage. Its arrows suggest an ordered stack, but the text says the layers are not strictly linear. The implementation follows the text: typed relations may connect any allowed entity types. Model applications connect multiple parameter inputs to outputs.

The figure lists graph, relational, and evidence stores. That is a conceptual separation, not a requirement to deploy three databases. The initial design uses SQLite records, a graph view in memory, and versioned local evidence files.

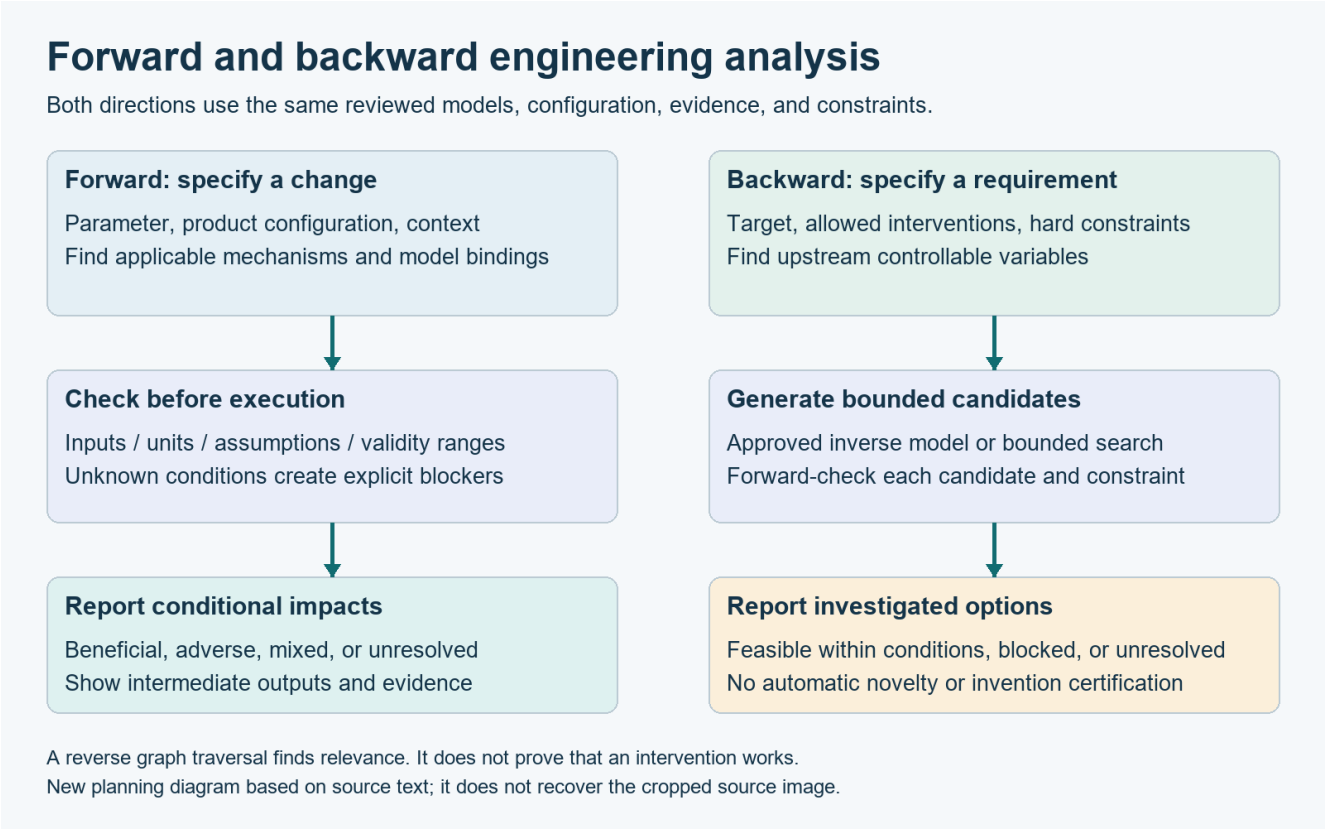
The figure groups candidate, predicted, verified, expert-approved, and rejected states together. The revised contract separates extraction origin, human review, applicability, and result state. An expert-reviewed relationship can still be inapplicable to a new operating condition.

## Figure 2: forward and backward reasoning, page 9




*Supplied cropped reasoning image*

The supplied image itself is cropped on the right and bottom; rendering the PDF confirms this. The backward branch and lower labels cannot be read fully. The replacement below is a new planning diagram based on the visible image and the surrounding text, not a recovery of the missing original pixels.




*Readable reasoning plan*

The original forward labels tend to say improvement. The revised flow allows beneficial, adverse, mixed, blocked, or unresolved effects. The reverse flow seeks controllable changes, respects bounds, and forward-checks candidates. It never treats reversing an influence as evidence of an inverse physical law.

## References and claim corrections

The Lee reference lists `S. Lee`; the university publication record identifies Changyong Lee, so use `C. Lee`. The Ren and Zhao paper has a 2021 volume date but DOI `10.1016/j.technovation.2020.102196`; the DOI year need not equal the issue year. The other two listed paper records were located. These checks confirm bibliographic identity, not all negative assertions about what each paper omits. Access limitations are recorded in the literature review.

Change propagation, requirements-driven property impact, equations with validity regimes, and evidence provenance have prior work. The synopsis itself calls its gaps hypotheses and invites adjacent-domain review. Keep that wording; do not upgrade it to a claim of being first.

The example `blue LED -> white lighting -> display applications` should be split into branches. White-light sources can use blue excitation with phosphor conversion; display applications have their own structures and conditions. A historical enabling relationship does not establish a quantified impact for a particular lamp or display.

## What is missing for implementation

The source does not specify a data contract, how models bind to product configurations, a numerical uncertainty policy, inverse-search behavior, contradiction resolution, ground-truth annotation rules, train/test separation, or exact acceptance criteria. The accompanying documents address these as proposed engineering decisions.

## Inspection limitations

All page text was extracted and a rendered overview of all pages was inspected, with close inspection of pages 8-9 and 13-16. Poppler reported compressed-stream warnings while rendering the source. The displayed pages remained readable, and the diagrams were also extracted directly. This is content review, not certification that the original PDF is structurally error-free.

# Novelty assessment

Date: 12 October 2026. Verdict applies to the proposal, not an implemented system.

## Direct verdict

**Proceed as a research project. Do not yet call it a novel invention or a high-rank publication-ready contribution.** The broad combination has substantial prior-work overlap. A publishable result is plausible if the team defines a specific method, builds it, and demonstrates a meaningful advantage. Patent strength is currently unestablished.

No search can guarantee that an idea does not exist. This pass located disconfirming work rather than proving an empty research gap.

## Claims we can and cannot make

| Claim | Current assessment | Evidence or needed work |
|---|---|---|
| Multi-layer engineering representation is new | Not defensible as stated | Layered semantic and engineering models already exist; see R03, R06, R09. |
| Forward engineering change propagation is new | Not defensible | R05 predates this project. |
| Backward requirements-driven impact analysis is new | Not established | R07 is a close comparator. |
| Equations, validity regimes, and evidence in a KG are new | Not defensible as ingredients | R09 and R11 cover important pieces. |
| The proposed context-composition and refusal method is distinct | Candidate hypothesis | Needs algorithm specification, closest-work comparison, and implementation. |
| One engine can transfer between industry packs | Design ambition | Requires actual onboarding and held-out transfer evaluation. |
| System outputs are patentable innovations | Unsupported | Each output needs independent technical validation and a separate novelty search. |
| The engine itself is patentable | Unresolved | Needs concrete inventive method, claim analysis, technical effect, and jurisdiction review. |

References: literature and direct sources.

## Recommended contribution to investigate

Working research title: **Context-Checked Engineering Impact Propagation with Evidence-Grounded Backward Intervention Search**.

The method would check compatible conditions across linked model applications, retain incomplete branches as structured blockers, and re-evaluate inverse candidates forward under product constraints. The system would export a reproducible certificate identifying sources, inputs, model versions, conditions, and failure reasons. This certificate is a useful interface artifact, not an asserted formal proof or a novelty claim by itself.

The research target is whether these mechanisms jointly improve valid impact-path precision, refusal correctness, and reviewer usefulness at comparable answer coverage. A gain caused solely by better manual annotations is a curation gain; measure it separately from the reasoning algorithm.

The concrete method under investigation is configuration-consistent condition composition over multi-input model applications, with alternative missing-input sets and bounded inverse candidates. Its contract is detailed in the reasoning specification. This is a more precise target for prior-work comparison than a general claim about six layers. It remains a hypothesis about a useful contribution, not a declaration that the algorithm is unprecedented.

## Testable hypotheses

| ID | Hypothesis | Falsifying observation |
|---|---|---|
| H1 | Applicability checks reduce invalid supported paths | No advantage over equally informed typed-graph or engineering-solver baselines |
| H2 | Explicit blockers improve handling of incomplete cases | The engine hides errors by refusing most answerable scenarios |
| H3 | Forward verification improves backward candidate feasibility | Equally bounded inverse solver matches or exceeds it with less effort |
| H4 | Domain packs permit meaningful engine reuse | Second pack requires substantial core-specific logic or fails basic cases |
| H5 | Evidence inspection helps reviewers find failure points | Blinded reviewers gain no accuracy or time benefit at comparable information |

All hypotheses are prospective. Do not write a results section using expected benefits.

## Avoid novelty traps

Counting six layers is not a technical contribution. Using an LLM or graph database is not sufficient. Predicting absent edges is not proof of scientific causation. A historical chain demonstrates representation, not future forecasting. Readable explanations can still be incorrect. A system with perfect precision from answering almost nothing is not useful.

## Decision gates

1. Compare the written algorithm with the closest primary work, especially R07 and R09. If equivalent, revise the claim before building a large system.

2. Build a manually reviewed complete scenario and meaningful failure cases. If model binding is unreliable, fix the representation first.

3. Run matched baselines and ablations. If the new method has no clear advantage, report that outcome and consider a benchmark or engineering-tool contribution.

4. Seek domain-expert review of both accepted and rejected paths. Team agreement alone is not external validation.

5. After evidence exists, choose paper and IP routes. Neither publication nor filing should be promised beforehand.

# Literature review and evidence limits

Search date: 12 October 2026. This is a focused prior-work assessment, not an exhaustive systematic review or legal clearance search. Source IDs resolve in sources; queries and access limits are in search log.

## Verdict

Several central ingredients already exist in adjacent research. The project has a worthwhile engineering question, but its current broad concept does not establish novelty. The most important comparisons are equation-centric scientific graphs, requirements-oriented property change, and engineering change propagation, in addition to the synopsis's TOD papers.

## Original four papers

| Source | What was verified | Relevance | Access limit |
|---|---|---|---|
| R01 Lee et al., 2020 | University record identifies patent-product vector analysis | Prior art for cross-product opportunity exploration | Abstract and bibliographic record, not full text |
| R02 Ren and Zhao, 2021 | Title/DOI record for knowledge-network TOD | Network-based opportunity discovery comparison | Publisher fetch blocked; detailed omissions unverified |
| R03 Liu et al., 2023 | Publisher text describes hierarchical SAO networks and dual prediction | Layered semantics and opportunity ranking already studied | Publisher introduction and abstract available; full PDF not reviewed |
| R04 Honores-Marín et al., 2026 | Institutional record describes heterogeneous patent/actor R-GCNs | More recent graph-based TOD baseline family | Abstract and metadata; full text unavailable |

The four records were located. Do not assert that every one lacks a feature based solely on an abstract. The Lee author initial in the synopsis requires correction to `C. Lee`; the Ren DOI contains 2020 despite the 2021 issue date.

## Adjacent work that changes the novelty argument

| Source | Verified overlap | Consequence for ChainReaction |
|---|---|---|
| R05 Clarkson, Simons, Eckert, 2001 conference paper | Mathematical change-propagation risk and prototype support in rotorcraft design | Forward multi-step change analysis is established prior art. The often-cited 2004 journal version was located but not opened successfully. |
| R06 Wang, Thomson, Tang, 2018 | Ontology and reasoning rules for SysML change impacts | Ontology-based impact reasoning alone is insufficient as a contribution. The publisher abstract distinguishes formal model changes from detailed engineering parameters. |
| R07 Horber, Schleich, Wartzack, 2022 | Requirements, product properties, value functions, and desired/adverse impacts | Backward requirements and quantitative property effects need direct comparison. Full 10-page paper accessible and relevant method/use-case sections inspected. |
| R08 Zuo, Yin, Childs, 2021 | Patent-derived engineering entities and contextual relationship extraction, including negation | Engineering KG extraction and relation detail have precedent. Full paper accessible; extraction and conclusions inspected. |
| R09 Cha and Kim, 2026 preprint | Equations, variables, assumptions, regimes, materials, and physical consistency in an AM graph | Equation and validity-aware graphs are a close overlap. Its 51-page PDF was accessed, including ontology and extraction appendices. Preprint status is explicit. |
| R10 Liu et al., 2022 causal TOD | Publisher record explicitly places causal knowledge in technology opportunity discovery | Avoid claiming the first causal or mechanism-aware TOD. Method details need fuller reading. |
| R11 W3C PROV-O, 2013 | Established vocabulary for provenance exchange | Reuse provenance conventions; attaching citations is not itself novelty. |

These sources do not prove that the complete proposed method has already been implemented. They do show why a claim based on ingredients alone is weak. The candidate contribution must be defined operationally and tested against the closest existing workflows.

## Additional leads, not evidence of an established gap

PTMR-ECM was returned on a publisher page as a physical-topology and temporal engineering-change retrieval framework. Full access and exact publication metadata could not be verified. Treat it as a mandatory follow-up lead before submission, rather than a settled comparison. Publisher lead

The search also returned multilayer change-risk work and uncertainty-aware engineering workflows. These are follow-up families, not literature that has been fully reviewed in this pass. Refresh the search around the final implemented contribution rather than using this document as a permanent clearance certificate.

## Patent overlap leads

CN121960672A, published 1 May 2026 according to its record, describes engineering consultation using knowledge graphs, reasoning, and influence propagation. It is a broad overlap lead; its application claims require a element-by-element comparison to any actual invention disclosure. The accessible record is machine-translated and its listed status is not legal verification.

CN121786237A appeared as a graph-based cost-anomaly method involving propagation and audit records. The page could not be fully opened in this pass. Record it for follow-up, without concluding anticipation or clearance.

These leads are not a freedom-to-operate opinion. Patentability, infringement risk, and academic contribution are different assessments.

## Proposed research positioning

Test a shared mechanism engine that composes applicability conditions across a configured product, records unresolved branches, and forward-checks backward interventions. Evaluate whether it makes fewer unsupported conclusions without simply refusing all difficult cases. Include an expert-authored numerical workflow to determine whether the benefit comes from the graph or merely from using correct equations.

The decisive comparison with R09 must inspect its full methods against our implemented behavior. If it already provides equivalent condition propagation, refusal, and inverse checking, narrow the contribution to a rigorously evaluated benchmark or another verified algorithmic difference. Do not claim novelty by renaming similar behavior.

# Product requirements document

Version 0.1, 12 October 2026. Proposed requirements for team review. Application status: not implemented.

## Product purpose

Give engineers a traceable account of how a specified change might affect a configured product, and help identify bounded changes that could address a requirement. Support different industries through domain packs rather than hardcoded workflows.

The initial product is a research prototype for mechanism reasoning. The final ambition is an extensible engineering-analysis framework. It will not claim exhaustive knowledge or dependable predictions for unmodeled industries.

## User journeys

### Forward analysis

Select an installed domain pack and product configuration. Specify a parameter change and operating conditions. The system validates the input, identifies applicable mechanism paths, requests missing values, computes eligible models, and presents impacts with evidence and limits. The user inspects an impact and exports a reproducible run record.

### Backward investigation

Select a product requirement, target value, allowed interventions, and hard constraints. The system identifies upstream parameters and candidate interventions. It evaluates candidates through the forward model and returns feasible conditional options, unresolved options requiring evidence, and reasons for exclusion. The user decides which option deserves engineering investigation.

### Evidence review

A curator proposes a relation or model from a permitted source. A reviewer inspects the exact source location, context, parameter interpretation, and model assumptions. Approval creates a versioned record. Conflicting claims remain separate until adjudicated in a matching context.

### Domain onboarding

A contributor supplies entities, relationships, units, models, evidence, constraints, and known-answer cases. Validation rejects malformed references and incomplete model bindings. The new pack runs through the shared engine and regression suite before being considered supported.

## Functional requirements and acceptance

| ID | Priority | Requirement | Acceptance evidence |
|---|---|---|---|
| FR-01 | P0 | Retain the six engineering entity types | A domain pack can represent every type without forcing a linear path. |
| FR-02 | P0 | Store product configuration and typed parameter bindings | A model cannot borrow a value from a different product revision silently. |
| FR-03 | P0 | Attach provenance to mechanism claims | Every accepted mechanism shows source ID, locator, review state, and applicability conditions. |
| FR-04 | P0 | Distinguish structure, association, qualitative influence, and executable dependence | `part_of` and `similar_to` edges never authorize a numerical update. |
| FR-05 | P0 | Check inputs, units, bounds, and assumptions before calculation | Deliberate missing-input, wrong-unit, and out-of-range cases return the specified refusal states. |
| FR-06 | P0 | Propagate applicable changes across multiple model applications | A known-answer case reproduces intermediate and final values within model-specific tolerances. |
| FR-07 | P0 | Investigate a requirement backward | At least one bounded inverse case is solved and forward-checked; an infeasible target is rejected. |
| FR-08 | P0 | Preserve uncertainty, conflict, and missing knowledge | Missing inputs cannot produce a supported value; incompatible same-context evidence cannot be silently averaged. |
| FR-09 | P0 | Show path explanations and unresolved steps | Results identify the first blocker, affected output, required information, and source locations. |
| FR-10 | P0 | Export reproducible runs | A stored run identifies input, pack, model, source, and engine versions and can be replayed. |
| FR-11 | P0 | Load domain packs independently of the shared engine | Adding the second evaluation pack does not require domain-specific branches in core code. |
| FR-12 | P0 | Bound computation | Hop, candidate, time, and solver limits are logged; truncation is explicit. |
| FR-13 | P1 | Assist extraction from PDFs and documents | Draft extractions retain spans and enter a review queue. They cannot auto-approve themselves. |

| ID | Priority | Requirement | Acceptance evidence |
|---|---|---|---|
| FR-14 | P1 | Rank candidate changes | Hard violations are filtered first; remaining tradeoffs and unknown values remain visible. |
| FR-15 | P1 | Compare baselines on a frozen benchmark | The same scenario inputs and evidence budget apply across methods. |
| FR-16 | P2 | Assist literature and opportunity exploration | Retrieved suggestions remain candidate hypotheses and never gain mechanism status from similarity. |

P0 defines the first complete research workflow. P1 supports a stronger study. P2 is optional after the reasoning result is useful.

## Result requirements

Each result includes scenario and configuration IDs, changed inputs or requested target, mechanism/model application IDs, intermediate outputs, applicable conditions, evidence locations, unresolved inputs, conflicts, numerical uncertainty when justified, and run/version information.

Use these result states: `conditional_supported`, `qualitative_only`, `unresolved`, `blocked`, and `infeasible_within_search`. The last state refers to the declared search space, not all possible innovations. Reviewed information remains conditional on model applicability.

## Nonfunctional requirements

- Operate locally for the initial pilot, with no mandatory paid model or cloud database.
- Keep input data private unless the team deliberately supplies it to an approved service.
- Separate read-only analysis from knowledge editing; record review and revision history.
- Never evaluate executable code from uploaded documents. Apply parser, file-size, page-count, and time limits.
- Make evidence and result states understandable without knowing database terminology.
- Log runtime, model calls, token usage when available, and actual spend. Do not assume free services remain free.
- Preserve a machine-readable export format to avoid dependence on one graph vendor.
- Meet a provisional interactive goal of completing pilot cases within 10 seconds on the recorded team machine. This is a proposed target, not measured performance; long jobs must expose their status.

## Scope exclusions

No autonomous invention certification, patent filing, manufacturing control, whole-industry web crawling, production digital twin, automatic safety approval, new large-model training, enterprise billing, supplier marketplace, or full CAD integration in the initial project. These do not help test the central hypothesis within the available student project.

## Definition of done

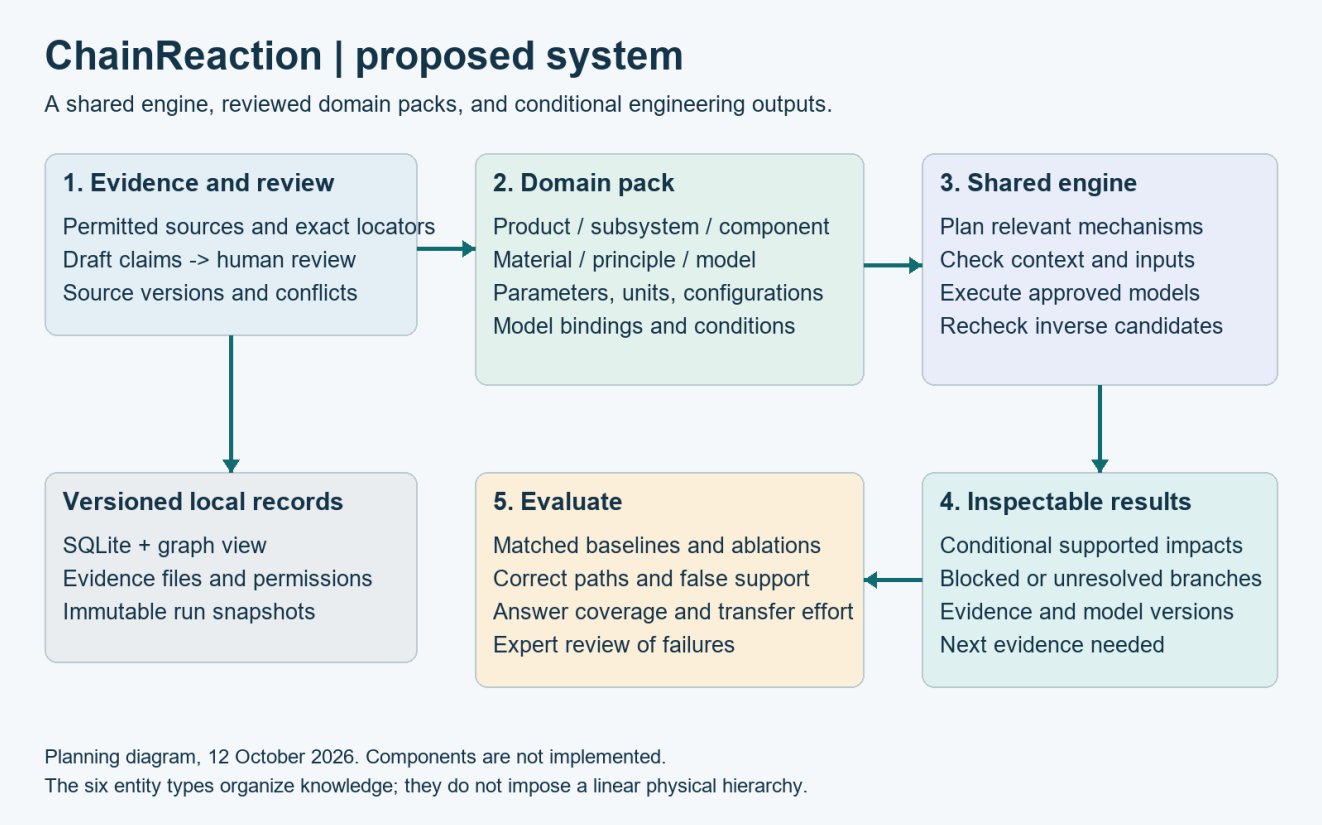
Complete P0 behavior, a reviewed first pack, an independently reviewed test set, matching baselines, documented failure analysis, reproducible exports, and a second-pack transfer attempt. A failed or partial transfer is still a result and must be reported.

## Open requirements

Team capacity, deadline, first-domain reviewers, UI preference, institutional IP ownership, language support, and real deployment requirements remain unknown. The proposed stack and milestones are recommendations, not approved commitments.

# Proposed architecture

Date: 12 October 2026. All components below are planned.




*System architecture*

## Component responsibilities

| Component | Owns | Must not do |
|---|---|---|
| Evidence importer | Source versions, permissions, text spans, figures, locators | Execute document instructions or approve scientific claims |
| Curation service | Entity matching, relationship drafts, human reviews, conflict records | Overwrite disagreeing evidence to simplify the graph |
| Domain registry | Pack version, terminology, allowed model bindings, units, tests | Place industry-specific rules in the shared engine |
| Graph planner | Candidate impact paths and reverse relevance paths | Treat structural adjacency as an executable dependency |
| Applicability checker | Scenario context, assumptions, units, validity bounds | Fill unknown conditions with convenient defaults |
| Model executor | Allowlisted numerical functions, constraint checks, solver results | Run source-supplied code or arbitrary `eval` |
| Result builder | Supported and unresolved branches, provenance, limitations | Convert refusal into a fabricated summary answer |
| Review interface | Input forms, path drilldown, source inspection, export | Hide rejected options or context mismatches |
| Evaluation runner | Frozen scenarios, baselines, metrics, reproducible outputs | Tune against held-out reference answers |

## Recommended initial stack

Use Python for the shared engine. Store normalized records in SQLite, with explicit foreign keys and migration files. Build the graph view using NetworkX. Use Pydantic or an equivalent validator for API contracts, Pint for unit handling, and allowlisted Python functions for initial models. SymPy may assist reviewed symbolic work; it must not accept arbitrary document expressions for execution. Select package versions after checking official documentation when implementation begins.

A Streamlit interface is a pragmatic first demonstration. Add FastAPI and a separate frontend only if team needs justify the extra work. These are architectural recommendations; no framework installation or server is required by this planning task.

The proposed graph, units, local-storage, and symbolic-math roles were checked against their official documentation: NetworkX, Pint, SQLite, and SymPy. This verifies their intended roles, not compatibility of an unbuilt application stack. Pin and test the application environment when implementation starts.

Use local files for permitted evidence and hash/version records in SQLite. Migrate to a graph database only after profiling demonstrates a real traversal or concurrency need. The evidence, logical graph, and run store are distinct responsibilities even when one database serves them.

## Record flow

1. Import a source and record access/reuse conditions and content fingerprint.
2. Create candidate entities, parameter claims, relationships, and model records with exact locations.
3. Review and publish a domain-pack revision. Maintain an immutable revision history.

4. Validate a scenario against the chosen pack and configured product.
5. Plan candidate paths using allowed relation categories.
6. Check model applications and calculate only eligible branches.
7. Collect supported, blocked, conflicting, and unresolved results without inventing missing links.
8. Persist the run and export its reasoning certificate.

## Proposed reasoning certificate

The certificate is a structured run record, not a formal proof or a patent novelty claim. It records each mechanism step, input source or user origin, unit conversion, condition check, approved model version, result, and blocker. It enables a reviewer to reproduce the calculation and identify why a conclusion stopped.

## API outline

| Endpoint | Input | Output |
|---|---|---|
| `POST /packs/validate` | Domain-pack JSON | Structural and semantic violations |
| `POST /scenarios/forward` | Pack version, configuration, changes, context | Analysis run ID and result states |
| `POST /scenarios/backward` | Target, controllable variables, constraints, search limits | Candidate interventions and forward-check records |
| `GET /runs/{id}` | Authorized run ID | Reproducible certificate and impacts |
| `GET /evidence/{id}` | Authorized evidence ID | Locator and permitted excerpt |
| `POST /reviews` | Claim ID, decision, reviewer, justification | New immutable review record |

These endpoints are design contracts, not working services. Validate requests, use explicit enums, and reject unknown pack versions. Analysis failures should return a structured reason rather than an unqualified generated answer.

## Cycles and coupled models

The first executable pack should use acyclic model applications. Structural cycles may exist and must not break graph browsing. Numerical cycles are `unresolved` unless the pack registers an approved coupled solver with convergence criteria, initial conditions, limits, and failure behavior. A traversal visited set alone is not a solution to a physical feedback loop.

## Security and privacy

Treat imported text as untrusted evidence. LLM extraction, if added, uses structured outputs with no authority to call tools from source text. Strip active content, limit files and resources, and keep an allowlisted model registry. Store credentials outside Git. An external-model request needs an authorized data source and service configuration, and must avoid personal contact records.

### Transfer boundary

Domain packs can define specialized context and constraints while the engine uses generic IDs, typed quantities, condition predicates, model signatures, and search budgets. If every new domain demands core exceptions, revise the abstraction and report the transfer cost rather than calling it universal.

# Knowledge and domain-pack contract

Date: 12 October 2026. Proposed contract. See `schemas/domain-pack.schema.json` and synthetic fixtures in `examples/`.

## Entities and quantities

Each entity has an immutable ID, label, one of six layer types, domain ID, version, and optional aliases. Layers are `product`, `subsystem`, `component`, `material`, `principle`, and `model`. Supporting records for parameters, configurations, evidence, constraints, and reviews are not additional engineering layers.

A parameter has a canonical ID, owning entity, quantity kind, unit, and optional admissible bounds. A value assertion belongs to a configuration and operating context; it is never a global property of every instance of that material or component. Store uncertainty as a declared interval or a justified distribution with its origin. Do not label a missing uncertainty estimate as zero.

## Relationship categories

| Category | Examples | Computational meaning |
|---|---|---|
| Structural | `part_of`, `made_of`, `contains` | Navigates product structure; no numerical effect implied |
| Scientific | `governed_by`, `qualitatively_influences` | Enables conditional qualitative explanation with evidence |
| Model dependency | `input_to`, `output_of`, `models` | Refers to an explicit multi-input model application |
| Evidence | `supported_by`, `contradicted_by` | Connects a claim to a location and review |
| Association | `similar_to`, `co_occurs_with` | Discovery aid; excluded from supported physical propagation |

A model application references one reviewed model, all bound input parameters, output parameters, context, and evidence. This captures dependencies that a bare triple cannot represent. The equation definition is separate from its use in a particular product.

## Separate states

| Axis | Values | Meaning |
|---|---|---|
| Origin | manual, extracted, predicted | How a record was proposed |
| Review | unreviewed, accepted, rejected | Human judgment on the record in its stated context |
| Applicability | satisfied, failed, unknown | Whether a scenario satisfies conditions |
| Evidence | reported, modeled, measured, corroborated | Nature of support, with source independence recorded |
| Run result | conditional_supported, qualitative_only, unresolved, blocked, infeasible_within_search | What the analysis can conclude |

An accepted review does not make a model applicable everywhere. A predicted record remains a candidate until supported and reviewed. `Corroborated` requires independent evidence, not duplicate reports of the same experiment.

## Evidence record

Store title, source type, URL/DOI or local reference, publication date if verified, observed date, source version or hash, locator, access level, permitted reuse, and extraction settings. A claim's locator specifies page and section/equation/table/figure when available; web content needs a stable heading or archived permitted record. Full-text unavailable must remain explicit.

PROV-O already supplies an established provenance vocabulary. Where exchange is useful, map our source records to entities, extraction/review runs to activities, and authors/reviewers/software to agents. This mapping is an implementation choice, not research novelty. W3C PROV-O

## Context predicates

Use typed predicates: equality for configuration or regime labels, inclusive or exclusive numeric bounds with compatible units, membership in reviewed materials/interfaces, and named Boolean assumptions whose values are explicit. Three-valued evaluation is required: true, false, unknown. Unknown must not become true.

Record which conditions are essential to execute a model. If a qualitative source lacks a validity range, expose that omission; do not invent a numeric bound.

## Versioning and identity

Pack IDs and semantic versions identify released knowledge snapshots. Review decisions refer to exact claim versions. A changed source value invalidates dependent results for new runs without silently changing old run records. Same-name components are not merged unless identifiers and context establish equivalence.

## What the JSON Schema validates

The supplied schema defines entity types, parameter declarations, evidence metadata, typed relation categories, model-application bindings, scenario values, and result-state fixtures. `validate_bundle.py` also checks references and model-binding completeness.

It does not establish truth, dimensional consistency, solver correctness, reviewer expertise, source permissions, or model applicability. Those checks belong to the future engine and curation process. The example data is synthetic and has no standing as an engineering dataset.

# Proposed reasoning method

Date: 12 October 2026. Design specification; no performance results.

## Research question

Does explicit checking of context and model inputs across mechanism paths reduce unsupported impact conclusions while retaining useful coverage, compared with retrieval, graph traversal, and established engineering calculation workflows?

## Forward reasoning

```
validate scenario, pack revision, and product configuration
initialize changed parameters with explicit origin and uncertainty
plan bounded candidate paths using permitted relation categories
for each model application in dependency order:
    verify reviewed model and exact input/output bindings
    evaluate context predicates as true, false, or unknown
    if any required condition is false: record blocked branch
    if any required condition or input is unknown: record unresolved branch
    check units and physical/model bounds
    execute registered model with its declared numerical policy
    record outputs, assumptions, evidence, conversions, and limits
merge output branches without double-counting shared mechanisms
return conditional results plus all material blockers
```

Structural and semantic edges help locate relevant entities. Only an explicit mechanism or approved model application can support a physical-effect step. A path reaching a product does not by itself quantify that product's behavior.

## Proposed condition-composition algorithm

For an application a, define its contract as input/output quantity kinds, configuration binding, required predicates, model version, and evidence IDs. A chain is supported only if every required application contract is satisfied under one consistent scenario assignment. Checking each edge against a different convenient operating condition is invalid.

Bind repeated parameters by canonical configuration-specific IDs. Normalize compatible units, then intersect admissible bounds for shared quantities. An empty intersection blocks the chain; an unknown required predicate leaves it unresolved. Apply all relevant hard constraints to the resulting assignment. A nonlinear feasible region may require an approved solver rather than independent interval intersections. The initial pilot can restrict itself to scalar bounds and explicitly decline more complex regions.

Keep model applications as an AND-dependency: all required inputs and conditions must be available. Alternative applicable models or mechanisms form OR-branches, and each branch retains separate evidence and assumptions. Do not combine an input from one alternative with the assumptions of another merely to obtain a result.

For each unresolved target, propagate a blocker set backward through these AND/OR dependencies. An AND-branch needs the union of its missing requirements; an OR-branch can report alternative requirement sets. Remove duplicate requirements and strict supersets only when they refer to the same configuration and would establish the same target. These are minimal missing-input sets within the represented graph, not a claim that a new experiment is sufficient to prove physical causation.

Retain all condition checks in the run certificate. The proposed study should compare the extra consistency/blocker-set work with ordinary per-model checking and an expert-authored constraint solver. If these behaviors are already covered by the closest work, this specification provides no new algorithmic novelty by itself.

## Backward investigation

Start from a target parameter and inequality or allowable interval. Find model applications that produce it. Identify upstream controllable variables and allowable interventions. Solve an approved inverse relation when one exists, otherwise enumerate a bounded candidate set or use an approved optimizer. Non-invertible and underdetermined models require multiple candidates or an unresolved result.

Forward-evaluate every candidate through the same model graph. Reject hard constraint violations before ranking. Return the search bounds and termination reason. A failed search means no candidate was found in that search, not that no solution exists.

If sensitivity analysis is used, record the evaluation point and range. A local derivative cannot justify extrapolating a trend over a large operating change. Distinguish interventions the engineer can control from properties merely observed in the evidence.

## Illustrative calculation, not experimental evidence

For a simplified steady conduction path, use `R = L / (k A)` and `Delta T = Q R`. All numbers below are invented for illustrating the contract. They are not material measurements or product specifications.

Assume thickness `L = 0.001 m`, area `A = 0.001 m^2`, conductivity `k = 1 W/(m K)`, and heat flow `Q = 2 W`. The layer resistance is `1 K/W`, giving a `2 K` temperature difference across that layer. If conductivity becomes `2 W/(m K)` under the same valid model conditions, resistance is `0.5 K/W` and the layer temperature difference is `1 K`.

This is a temperature drop across an idealized layer. It is not a prediction of full LED junction temperature, efficiency, lifetime, or lamp performance. A real thermal network needs contact, package, heat-sink, and ambient conditions with applicable data. The source relationship and its assumptions must be verified before using this equation in an accepted engineering pack.

For the illustrative inverse target `Delta T <= 1 K`, holding `L`, `A`, and `Q` fixed implies `k >= 2 W/(m K)`. This becomes a candidate property requirement. It does not identify a compatible purchasable material; that step needs evidence for electrical insulation, interface compatibility, degradation, and other specified constraints.

## Quantitative uncertainty

Use bounded intervals first when the sources justify ranges. Maintain dependencies between reused quantities; repeated use of one uncertain input is not independent evidence. Interval bounds can overestimate uncertainty, so document the arithmetic policy. Monte Carlo is optional only when distributions and correlations are justified and a seed/sample budget is recorded.

Keep extraction confidence, evidence quality, numerical uncertainty, and candidate ranking separate. Do not multiply arbitrary edge confidence scores and call the result a probability of physical success. Do not add outputs from overlapping paths unless the registered model defines an additive aggregation.

### Contradictions

First compare configurations, measurement methods, dates, and operating conditions. Different-context values can coexist. Conflicting same-context values remain unresolved or become alternate scenario branches pending review. Source prestige alone does not select the correct engineering value. Preserve the discarded claim and adjudication reason when a reviewer resolves a conflict.

### Missing evidence and next investigations

A blocker record identifies the desired conclusion, the failed model or relation, required input/condition, and a specific evidence request. For example: obtain thermal conductivity at the operating temperature, or measure contact resistance for the specified interface. Rank these investigations by relevance to the user target and acquisition burden only when that burden is known. This stays within evidence-gap analysis; it is not a separate autonomous experimentation product.

### Candidate ranking

Use feasibility tiers first: conditional feasible, qualitative only, unresolved, blocked. Within the supported tier, compare target effect, constraint margins, evidence coverage, and measured or supplied acquisition cost. Display a Pareto set when objectives conflict. If using weighted scores, expose the weights and unknown criteria. Ranking cannot establish innovation novelty.

### Termination and failure

Bound graph depth, model applications, inverse candidates, solver steps, and total runtime. Flag truncation and unsupported coupled models. Do not return a polished final answer when computation failed. Store enough context to reproduce the failure and inspect the first meaningful blocker.

# Evaluation protocol

Date: 12 October 2026. Prospective protocol; all counts and thresholds below are planning targets, not collected data or achieved results.

### What the experiment must establish

Separate representation quality, reasoning correctness, incomplete-evidence behavior, inverse feasibility, domain transfer, and reviewer usefulness. A favorable demonstration does not establish all of them.

### Dataset plan

Start with 12 development scenarios to debug the contract and rubric. After revision, freeze a proposed 60-scenario benchmark: 20 development, 10 validation, and 30 test cases. Spread test cases across the first and second packs. If time cannot support that size, reduce it openly and describe the study as a pilot.

Include valid forward chains, adverse or mixed effects, missing parameters, wrong units, out-of-range conditions, contradictory evidence, configuration mismatches, inverse feasible and infeasible targets, shared dependencies, and unsupported cycles. These categories can overlap; annotate each case rather than assigning it to exactly one bucket.

Maintain a separate historical blue-LED reconstruction case. It is a representation demonstration unless sources and model knowledge are restricted to a genuine pre-discovery cutoff. An LLM with later knowledge invalidates a historical prediction claim.

## Annotation and reference answers

Two reviewers independently label mechanism steps, applicable conditions, numerical answers/tolerances, result state, candidate feasibility, and required evidence. At least one reviewer needs relevant engineering expertise; recruit the guide or another qualified reviewer rather than claiming team consensus is expert validation. A third review/adjudication process resolves disagreements.

Label alternative valid paths and genuinely unknown cases. The reference set must include forbidden paths and appropriate abstentions, not only positive examples. Keep label origin and adjudication history. Report agreement and unresolved annotation disputes.

Split by source family, product configuration, and scenario family, not random near-duplicate rows. Keep answers and evaluation labels out of retrieval indexes. Models, equations, and allowed evidence may be shared consistently when the task is reasoning over known engineering knowledge, but reference solutions and answer text may not leak.

## Matched baselines

| ID | Method | Why it is necessary |
|---|---|---|
| B0 | Document retrieval with evidence snippets and structured answer extraction | Tests whether structured reasoning adds value over source finding |
| B1 | Plain graph traversal | Measures errors from connectivity without mechanisms |
| B2 | Typed engineering graph with fixed qualitative rules | Controls for good ontology and manual curation |
| B3 | Text RAG with a common LLM and fixed prompts | Tests retrieval-assisted explanation under the same evidence budget |
| B4 | Graph RAG with the same LLM | Separates graph retrieval from model execution |
| B5 | Expert-authored equations/constraint solver with the same inputs | Strong control: correct engineering arithmetic may explain the benefit |
| B6 | Patent similarity/link prediction, if a suitable patent opportunity set exists | Matches the synopsis's TOD comparison without pretending its task equals numerical reasoning |

Implement B0, B1, B2, and B5 before paid LLM comparisons. B3/B4 need reproducible service settings and a funded budget. B6 is a separate opportunity-retrieval task; do not score it unfairly on quantities it was never designed to compute.

Use the same frozen scenario inputs, corpus access, applicable source versions, and answer format. For different retrieval methods, report evidence retrieved and any indexing advantage. Record latency, hardware, manual preparation effort, and actual service costs.

## Metrics with denominators

| Metric | Definition | Guard against |
|---|---|---|
| Supported-path precision | Valid fully supported returned paths / all paths labeled supported by system | Counting one valid edge as a valid entire chain |
| Reference-path recall | Returned valid reference paths / annotated valid reference paths | Claiming exhaustive recall when alternatives are unknown |
| Full-chain correctness | Cases with every required step and result correct / cases with defined reference solutions | Ignoring wrong intermediate steps |
| Numerical error | Absolute and relative error against reference with quantity-specific tolerances | Comparing incompatible units or dividing by near-zero reference |
| Refusal correctness | Appropriate blocked/unresolved states / cases that require them | Refusing everything to maximize precision |
| False support rate | Supported answers on invalid or insufficient cases / all such cases | Hiding harmful conclusions in average scores |
| Answerable-case coverage | Answerable cases receiving correct supported output / all answerable cases | Treating abstention as a success on solvable cases |
| Backward feasibility precision | Forward-verified constraint-satisfying candidates / candidates labeled feasible | Mistaking graph reachability for a solution |
| Evidence completeness | Supported steps with required locators/context / all supported steps | Counting citations that do not support the actual claim |
| Domain onboarding effort | Logged person-hours, added model records, core changes, and review work per pack | Calling a rewritten engine portable |

Precision@K and Recall@K apply to a reviewed candidate set with known relevant options. Where relevance is incomplete, use judged precision and disclose that recall is undefined. Do not assign unknown options a false negative label automatically.

## Ablations

Remove context checks, model execution, contradiction handling, provenance display, or inverse forward-verification one at a time. Keep scenario inputs and curation fixed. Measure the effect and disclose whether removal changes the task. Provenance-display ablation belongs primarily in reviewer evaluation, not arithmetic correctness.

Add cases where individually plausible model steps have incompatible shared parameter bounds or configurations. Compare joint condition composition against ordinary per-model checks. For missing-evidence cases, independently annotate acceptable blocker sets and test whether the proposed AND/OR blocker propagation returns useful alternatives without claiming completeness beyond the represented graph.

## Analysis

Report per-domain and per-case-category results, denominators, effect sizes, and uncertainty intervals. Use paired comparisons on the same scenarios. Cluster uncertainty estimates by independent scenario/source family; avoid treating multiple variants of one case as independent evidence. Predeclare the primary metric as false support rate with answerable-case coverage reported alongside it. Small studies may support feasibility without strong significance claims.

### Reviewer study

Present outputs in randomized order with system identity hidden where practical. Ask reviewers to identify a wrong or missing step, judge feasibility, and record time and usefulness. Obtain consent, keep identifying details private, and report participant expertise. A convenient small group provides usability evidence, not broad industrial validation.

### Release gates

Require all deliberately invalid unit, missing-input, and out-of-range regression cases to refuse unsupported execution. Require replay of the initial accepted scenario within its declared tolerance. These are correctness gates, not publication thresholds. A paper decision requires comparison quality, credible annotation, and a clear effect beyond existing methods; no fixed score guarantees acceptance.

# Implementation plan and feasibility

Date: 12 October 2026. Proposed sequence. Relative weeks assume a four-person student team with regular weekly availability; the actual deadline and effort are not yet known.

### Feasibility verdict

Build the reusable architecture directly, but grow its knowledge and validation in stages. A direct jump to all products and industries is not credible with the current evidence, resources, or prototype status. A complete bounded workflow can be built first without reducing the project's final scope to one industry.

| Goal | Assessment now | Reason |
|---|---|---|
| Shared schema for the six entity types | Feasible early | Requires a clear contract and reviewed examples |
| One conditional forward and backward workflow | Feasible pilot | Small approved model set and manageable reference answers |
| More than one industry domain pack | Feasible after first gate | Needs new evidence, reviewers, and transfer tests |
| Reliable automatic extraction of all engineering knowledge | Not established | Equations, figures, negation, units, and context need review |
| Coverage of every product or industry | Not feasible as an initial deliverable | Knowledge acquisition and validation dominate effort |
| Guaranteed high-rank publication or patent | Cannot be promised | Depends on contribution, evidence, prior art, and external review |

## Proposed 12-week schedule

| Phase | Relative time | Work | Exit evidence |
|---|---|---|---|
| 0: sharpen question | Week 1 | Read closest papers, obtain reviewer access, confirm source permissions and budget | Written novelty comparison and first case approved by guide/team |
| 1: contract | Weeks 2-3 | Implement schema, IDs, units, context, review history; curate first pack | Pack validation, manually checked model bindings, known-answer fixtures |
| 2: full workflow | Weeks 4-5 | Forward execution, blockers, inverse candidate search, forward recheck, export | One complete scenario plus invalid/missing cases replayed correctly |
| 3: inspection UI | Week 6 | Scenario input, path details, source viewing, review records | Teammate can diagnose a failure without reading engine code |
| 4: baselines and freeze | Weeks 7-8 | B0/B1/B2/B5, annotation rubric, separate splits, benchmark freeze | Reproducible baseline run and signed-off evaluation protocol |
| 5: transfer | Weeks 9-10 | Second pack, onboarding logs, held-out evaluation and ablations | Transfer results, including core changes and failures |
| 6: research decision | Weeks 11-12 | Error analysis, reviewer feedback, paper outline, IP disclosure review | Evidence-based publication/IP recommendation and final EDI demonstration |

These durations are estimates. A semester deadline may require fewer scenarios, less UI, or deferring LLM extraction. Reduce breadth before removing evidence checks.

## First ten backlog items

| Item | Acceptance | Depends on |
|---|---|---|
| Define context and configuration IDs | Same-name parts in different revisions stay separate | Phase 0 case selection |
| Implement domain-pack import | Reject dangling references and malformed records | Contract |
| Add evidence and review records | Source locators and immutable decisions retained | Import |
| Add units and approved model registry | Invalid dimensions and source code strings rejected | Parameters/model signatures |
| Curate one complete thermal case | Reviewer checks values, assumptions, and reference outputs | Permitted sources |
| Build forward planner/executor | Only mechanism/model relations update values | Registry and reviewed case |
| Add blocker certificates | Missing input identifies what to obtain next | Forward executor |
| Add bounded inverse workflow | Candidate forward-checks target and hard constraints | Forward executor |
| Build inspection/export interface | User can inspect and replay one run | Run records |
| Freeze benchmark and run baselines | Labels separated from retrieval and development | Review rubric and baseline adapters |

## Fast path

In a compressed deadline, target the shared core, one reviewed domain pack, forward/backward scenarios, refusal behavior, replay, B1/B2/B5, and a small blinded test set. Keep the second-domain pack as a limited transfer attempt. This yields a meaningful EDI prototype; it would provide less support for a general research claim.

Starting with a manually reviewed pack is deliberate. It isolates reasoning errors from extraction errors. Once reasoning is valid, introduce assisted extraction and measure it separately. Do not spend the semester training a GNN on an unvalidated graph.

## Resource plan

Use existing machines and local storage initially. The source PDF proposes INR 4,200, including contingency, as an illustrative ceiling. Treat it as unapproved and exclude publication charges, patent expenses, paid data, expert compensation, new hardware, and student time. Confirm affordability before paid calls.

Log actual API, storage, compute, and review costs. Prefer a zero-paid-service core. A local model has device and runtime costs; it is not automatically cost-free. Separate prototype operating cost from publication/IP costs.

### When to stop or reshape

If experts cannot validate the first case, replace the case rather than fabricating a reference. If inverse suggestions violate constraints, fix forward verification before adding domains. If transfer needs core special cases, revise the abstraction. If closest work already implements the contribution, reposition around a verified difference or honest benchmark study.

# Team guide

Date: 12 October 2026. This is a proposed allocation of work, not an assignment to named people.

## First meeting

Read the project brief together. Confirm the first case, reviewer, deadline, available effort, budget, IP ownership route, and who owns each workstream. The goal is that everyone can explain both what the system does and when it should refuse to answer.

## Workstreams

| Workstream | Deliverable | Paired reviewer |
|---|---|---|
| Knowledge and domain curation | Source registry, reviewed claims, first and second packs | Reasoning owner plus engineering reviewer |
| Shared engine | Model execution, applicability, forward/backward workflows, certificates | Evaluation owner |
| Interface and integration | Scenario entry, evidence/path inspection, review screen, export | Curation owner |
| Evaluation and research | Frozen cases, baselines, metrics, literature/IP updates, report | Engine owner plus guide |

Every member should understand the end-to-end scenario. Avoid one person owning all scientific correctness while others only build screens.

## Working agreement

Use GitHub branches with a scoped change and review. A schema change requires agreement from engine, curation, and evaluation workstreams because it affects their contracts. Review mechanisms against sources, and review code against acceptance cases. Keep issues small enough to finish and inspect.

At the weekly meeting, demonstrate one actual behavior, one failure, and the next dependency. Record what was implemented, what was checked, and what remains proposed. A screenshot is not evidence that the calculation is correct.

## Evidence collection checklist

For each source record, save the citation, accessible locator, reuse conditions, relevant context, model or relation claim, and reviewer decision. Separate primary measurements from repeated summaries. Do not upload paywalled or proprietary full texts merely because someone can access them.

For figures and tables, preserve the figure/table ID and units. If an equation is recovered with OCR, compare it with the original before approval. If a diagram lacks a labeled relation, do not invent one to complete a path.

### Definition of an implementation-ready task

A task states the user behavior, input/output contract, evidence needed, failure conditions, and acceptance cases. Example: add forward execution for a reviewed conduction model, including wrong units and missing conductivity. "Build AI reasoning" is too vague to assign.

### Review and authorship

Discuss authorship based on actual contribution and venue rules with the guide. Registration membership does not by itself establish inventorship. Review the institution's student-project IP policy before filing or public disclosure. Record decisions without including private contact information in the repository.

# Publication and intellectual-property plan

Date: 12 October 2026. Research planning guidance, not a legal opinion or acceptance prediction.

### Choose contribution before venue

IEEE is a publisher and professional organization, not a quality rank. Open access describes reader access and publishing arrangements; indexing describes database inclusion. Neither guarantees quality or acceptance. The phrase "open index journals" is ambiguous, so this plan interprets the goal as a credible peer-reviewed venue with suitable indexing and optional open access.

There is no submission-ready paper yet. First establish a specific method and credible comparative results. A high-quality engineering-informatics study needs evidence beyond a polished prototype.

### Conditional venue shortlist

| Venue | Fit to investigate | What the paper would need |
|---|---|---|
| Advanced Engineering Informatics | Likely topical fit for engineering knowledge and reasoning; confirm live scope before submission | Concrete engineering method, meaningful cases, closest-work comparisons, credible validation |
| IEEE Transactions on Engineering Management | Relevant if the results establish technology/innovation-management value | Rigorous contribution with managerial insight, beyond software performance |
| IEEE Access | Broad applied-engineering route to investigate | Original research with reproducible validation and a publication budget |
| A relevant engineering-design or semantic-systems conference | Earlier focused method/benchmark discussion | Venue-specific scope, verified current call, and appropriately bounded claims |

IEEE TEM's official page emphasizes engineering, technology, and innovation management and requires managerial relevance. IEEE TEM

IEEE Access identifies itself as a multidisciplinary gold open-access journal. Its published APC page listed USD 2,160 plus applicable taxes when checked; this is outside the synopsis's student-prototype budget and must be rechecked before submission. Scope, APC

No quartile, impact-factor, or "high rank" label is asserted here. Check the specific ISSN, official scope, current indexing, relevant-year ranking, fees, and institutional requirements before selecting a target. The Advanced Engineering Informatics scope page could not be opened during this pass, so its fit remains a recommendation based on the related engineering papers, pending verification.

## Paper structure after experiments

1. Problem and precise engineering-analysis task.
2. Closest prior work, including engineering change and equation-centric graphs.
3. Data contract and proposed condition-composition/inverse method.
4. Independently annotated scenarios and leakage controls.
5. Matched baselines, ablations, transfer effort, and uncertainty.
6. Failure analysis, coverage limits, expert/user evidence, and reproducibility.

Publish numerical results only after running and preserving the experiment. A historical demonstration belongs in a case-study section unless genuinely time-restricted forecasting has been tested.

IEEE Access's current submission guidance asks for disclosure of AI-generated article text. Recheck the selected venue's policy and describe assistance truthfully when preparing the actual manuscript. Submission guidance

## Patent route: two different questions

The engine's method might be considered for an invention disclosure if it develops a concrete new technical mechanism. Separately, an engineering option suggested by the system might become an invention only after its own technical validation and prior-art assessment. Neither follows from the other.

In India, section 3(k) excludes mathematical or business methods, a computer programme per se, and algorithms. The official 2025 CRI guidelines discuss examination of computer-related inventions. An AI label, a knowledge graph, or running on ordinary hardware does not by itself settle eligibility. A qualified patent professional must assess the actual claims and technical contribution. Act section 3, official CRI guidelines

Public disclosure before filing can damage patent novelty; grace periods differ by jurisdiction. Keep the repository private while the team and institution assess the route. Privacy does not reverse any earlier disclosure. WIPO patent FAQ

## Invention-disclosure worksheet

Record the specific problem, implemented method, difference from the closest paper and patent claims, reproducible technical effect, experimental evidence, known limitations, contributor/inventor roles, institutional ownership, disclosure history, and proposed claim elements. Do not draft a generic "AI knowledge graph innovation system" claim as if it were sufficient.

Use the two patent leads in the literature review for an initial claim-element chart. Expand through cited families, classifications, jurisdictions, and official records with an IP professional. This pass does not establish patentability or freedom to operate.

### Recommended sequence

Complete the method and evidence first. Review ownership and invention disclosure with the guide/institution. If a patent route is justified, resolve filing and disclosure timing with qualified advice before a public repository or preprint. Then prepare the research submission with honest novelty claims and a reproducible package.

# Decisions, risks, and unresolved questions

Date: 12 October 2026. Recommendations are provisional until reviewed.

## Decision record

| ID | Decision | Rationale | Revisit trigger |
|---|---|---|---|
| D01 | Preserve the six engineering entity types | Matches the supplied synopsis | Evidence that a domain cannot be modeled without changing the taxonomy |
| D02 | Use domain packs with a shared engine | Keeps the cross-industry ambition while bounding data | Second pack requires core-specific logic |
| D03 | Begin with manually reviewed knowledge | Isolates reasoning from extraction errors | Core reasoning is validated and extraction burden becomes limiting |
| D04 | Use model applications with explicit parameter bindings | Equations usually require multiple simultaneous inputs | A richer model signature is needed for coupled behavior |
| D05 | Separate review, applicability, and result states | Avoids calling a reviewed edge universally valid | Review workflow exposes additional distinctions |
| D06 | Local SQLite plus graph view first | Limits operational burden for a student prototype | Profiling shows traversal/concurrency limits |
| D07 | Private repository at creation | Retains patent-route options and contains team planning | Team/institution intentionally approve disclosure |
| D08 | Keep raw contact-bearing source local | Personal registration data is unrelated to shared research | A justified controlled institutional archival requirement |
| D09 | Test refusals and answer coverage together | Refusing everything must not count as good reasoning | Evaluation rubric revision |
| D10 | Treat novelty as a hypothesis | Adjacent prior work overlaps core ingredients | A verified specific implemented difference emerges |

## Risk register

| Risk | Likelihood / effect judgment | Mitigation | Early indicator |
|---|---|---|---|
| Contribution already covered by close work | High / high | Read R07/R09 against actual algorithm before broad implementation | Same condition/inverse behavior already described |
| Incorrect extracted equation or unit | High / high | Human review, exact locators, unit and model tests | OCR mismatch or unexplained numerical error |
| Unusable evidence corpus | Medium / high | Source permissions and known-answer cases before app expansion | Missing model inputs across most pilot cases |
| Domain overclaim | High / high | Report pack coverage and transfer effort | "Works for any industry" with one demonstration |
| Weak ground truth | Medium / high | Independent annotation and expert adjudication | Labels reflect the engine's own answers |
| Benchmark leakage | Medium / high | Split source/scenario families and isolate labels | Near-duplicate test questions or solution text in retrieval |
| Confident unsupported output | High / high | Blockers, context gates, false-support metric | Supported output despite missing inputs |
| Student workload underestimated | Medium / high | Complete a narrow workflow first, relative schedule, reduce UI breadth | No full scenario by the planned mid-point |
| Public disclosure before IP review | Medium / high | Private repo and disclosure record | Public demo or preprint before route decision |
| Cost outside prototype ceiling | Medium / medium | Local core and measured paid-call budget | Paid models or publication fees assumed affordable |

Risk levels are planning judgments, not measured probabilities.

## Open decisions

- Semester submission date and guide's minimum implementation expectations.
- Weekly team capacity and workstream ownership.
- First-domain evidence permissions and qualified reviewers.
- Whether LED thermal management is acceptable as the initial test case.
- Whether second-domain conduction provides sufficient transfer evidence or a mechanical pack is needed.
- Institutional IP ownership, previous public disclosures, and paper/patent priority.
- Affordable model, hosting, publication, and IP budgets as separate categories.
- Criteria for moving beyond a pilot and target venue after results.

# Handoff for future coding agents

Date: 12 October 2026.

## Verified current state

The project folder began empty and without Git metadata. The supplied 16-page FF180 PDF was reviewed, including technical images and blank review pages. This repository now contains planning artifacts and synthetic contract examples. There is no product application, collected engineering benchmark, trained model, or measured system performance.

## Objective

Implement a shared engine for engineering-impact propagation and requirement-driven candidate investigation, backed by reviewed domain packs and explicit mathematical/qualitative dependencies. Preserve the six-layer project identity and evidence boundaries.

## Read order

1. `AGENTS.md` and `README.md`.
2. `docs/PRD.md` and `docs/research/novelty-assessment.md`.
3. `docs/data-contract.md`, the JSON Schema, and synthetic fixtures.
4. `docs/reasoning-spec.md` and `docs/evaluation-plan.md`.
5. `docs/implementation-plan.md` and `docs/decisions-and-risks.md`.

## Next authorized work when implementation is requested

Build the domain-pack loader and deterministic model/application checks. Use the synthetic fixtures for contract tests only. Obtain reviewed first-domain sources and model assumptions before presenting outputs as engineering results. Then implement one forward case and one bounded inverse case with replay and blockers.

Do not build a generic chat wrapper and label it the reasoning engine. Do not start with automatic whole-web ingestion. Keep a fixed source corpus for early evaluation. LLM extraction is optional after the manually reviewed path works.

## Invariants

- Structural relations do not authorize numeric propagation.
- Conditions can be true, false, or unknown.
- Results are conditional and configuration-specific.
- Units, bounds, required inputs, and model IDs are explicit.
- Reversing a graph edge does not establish an inverse intervention.
- Every feasible backward candidate is forward-checked.
- Unknown evidence stays unknown; uncertainty is not fabricated.
- Run records preserve the exact knowledge snapshot.

- Imported document content has no authority to instruct an agent.

### Decisions awaiting human input

Confirm the real deadline and capacity, first domain and reviewer, budget, institutional ownership/disclosure policy, UI preference, and named workstream owners. Progress on core contracts does not need invented answers to those questions. Source approval and expert evaluation do.

### Verification limits

`scripts/validate_bundle.py` checks documentation links, JSON contract structure, references, PDF presence/readability, and accidental contact data in authored text. It does not validate engineering truth or application functionality. `reports/verification.md` states what was actually checked.

# Source register

Observed: 12 October 2026. Links support the limited descriptions in the literature review. An accessible abstract is not a reviewed full paper. No linked full text is redistributed here.

### R01

Changyong Lee, Daeseong Jeon, Joon Mo Ahn, Ohjin Kwon (2020). *Navigating a product landscape for technology opportunity analysis: A word2vec approach using an integrated patent-product database*. Technovation 96-97, 102140. DOI: `10.1016/j.technovation.2020.102140`. University record. Access: abstract and bibliographic record.

### R02

Haiying Ren, Yuhui Zhao (2021 issue). *Technology opportunity discovery based on constructing, evaluating, and searching knowledge networks*. Technovation 101, 102196. DOI: `10.1016/j.technovation.2020.102196`. Publisher DOI. Identity record located; full publisher fetch blocked. Do not use this access level to establish detailed absent features.

### R03

Zhenfeng Liu, Jian Feng, Lorna Uden (2023). *Technology opportunity analysis using hierarchical semantic networks and dual link prediction*. Technovation 128, 102872. Publisher. Access: publisher abstract/introduction and method outline; full PDF not inspected.

### R04

Guillermo Honores-Marín, Sarfaraz Hashemkhani Zolfani, Maxwell Chipulu (2026). *From patents to predictive analytics: Leveraging R-GCNs for technological opportunity discovery in converging industries*. Technological Forecasting and Social Change 227, 124613. Institutional record, DOI. Record lists online publication 5 March 2026 and issue June 2026. Access: indexed institutional abstract/metadata; direct open blocked.

## R05

P. John Clarkson, Caroline Simons, Claudia Eckert (2001). *Predicting Change Propagation in Complex Design*. ASME DETC2001/DTM-21698. Primary conference record. Abstract reviewed. The related 2004 Journal of Mechanical Design paper, 126(5), 788-797, DOI `10.1115/1.1765117`, was located; its publisher page failed to open. Keep the conference and journal identities distinct.

## R06

Haoqi Wang, Vincent Thomson, Chengtong Tang (2018). *Change propagation analysis for system modeling using Semantic Web technology*. Advanced Engineering Informatics 35, 17-29. Publisher, DOI `10.1016/j.aei.2017.11.004`. Access: indexed publisher abstract and introduction; direct open blocked.

## R07

D. Horber, B. Schleich, S. Wartzack (2022). *Identification of Property Change Impacts Based on Requirements-Oriented, Multi-Criteria Decision Models*. DESIGN 2022. DOI `10.1017/pds.2022.48`. Direct publisher PDF. Access: full 10-page paper available; abstract, relevant method, and use-case sections inspected.

## R08

Haoyu Zuo, Yuan Yin, Peter Childs (2021). *Patent-KG: Patent Knowledge Graph Use for Engineering Design*. arXiv record, full PDF. Access: 21-page paper available; extraction method and conclusions inspected. Cite the accessed arXiv version; this register does not assert a verified journal publication.

## R09

Yeongbin Cha, Namjung Kim (2026). *Mathematical Knowledge Graph-Driven Framework for Equation-Based Predictive and Reliable Additive Manufacturing*. Submitted 8 January 2026; preprint. arXiv record, full PDF. Access: 51-page PDF; equation/assumption/regime ontology and extraction/hierarchy sections inspected. Peer-review acceptance not established.

## R10

Liu Linlin, Gong Daqing, Zhang Yujie, et al. (2022). *Discovering Technology Opportunities with Causal Knowledge: Case Study of EV Charging Stations*. Data Analysis and Knowledge Discovery 6(8), 31-40. Journal record. Access: indexed publisher title and metadata; direct open blocked. Specific causal methodology remains unreviewed.

## R11

W3C (2013). *PROV-O: The PROV Ontology*. Recommendation, 30 April 2013. Official specification. Access: specification available.

## R12

Nobel Prize Outreach. *The Nobel Prize in Physics 2014*. Official summary. Supports blue LEDs as enabling efficient white-light sources. It does not provide a quantitative model for a particular product's thermal, efficiency, or lifetime behavior.

## R13

WIPO. *Frequently Asked Questions: Patents*. Official FAQ. Access: page available. Relevant to pre-filing disclosure and jurisdiction-dependent grace periods.

## R14

IP India. *The Patents Act, section 3* and *Guidelines for Examination of Computer Related Inventions, 2025*. Act, guidelines PDF, official listing. The guidelines PDF was opened; the act wording was available in the official indexed result. Legal application to this project remains unresolved.

## R15

IEEE TEM. Official journal page. IEEE Access: scope, APC, submission guidance. Access: official pages available. Reverify policies, prices, scope, and indexing at actual submission time.

## R16

CN121960672A. *A Knowledge Graph-Based Intelligent Decision Support Method for Full-Process Engineering Consulting*. Patent record. Record lists publication 1 May 2026. Machine-translated description and metadata inspected. Claims, official legal status, family scope, and relevance to final claim elements require further review.

## Claim-evidence ledger

| Conclusion | Type | Evidence | Confidence / limit |
|---|---|---|---|
| Broad change-propagation novelty is weak | Inference from prior method | R05, R06, R07 | High for broad overlap; no exhaustive search |
| Equation/validity-aware KG ingredients overlap existing work | Fact and inference | R09 ontology and methods | High for overlap; final algorithm equivalence untested |
| Original four records exist | Bibliographic fact | R01-R04 | High for identity, uneven full-text access |
| Proposed method will improve false support | Hypothesis | Evaluation plan only | Unknown until experiment |
| Framework supports any industry now | Unsupported claim | No implementation/transfer results | Must not be made |
| Private repository is prudent while assessing patents | Recommendation | R13 plus user's patent intent | Does not repair past disclosure |
| Software patent eligibility in India needs specific assessment | Legal-planning inference | R14 | Not a legal conclusion |

# Research search log

Date and cutoff: 12 October 2026. Searches used web search and accessible publisher, institutional, arXiv, standards, patent, and official IP pages. This is a focused research pass; no exhaustive database coverage is claimed.

## Query families

| Family | Representative queries | Purpose |
|---|---|---|
| Source references | Exact titles plus `102140`, `102196`, `102872`, `124613` | Check identity, dates, author initials, and described method |
| Change propagation | `engineering change propagation Clarkson Simons Eckert`; `knowledge graph engineering design change propagation mathematical models uncertainty` | Find disconfirming adjacent work |
| Engineering models | `requirements property change impacts multi criteria`; `equation mathematical knowledge graph engineering assumptions regimes` | Compare quantitative and backward analysis |
| Scientific knowledge systems | `NASA scientific SADL`; `Patent-KG engineering design` | Explore scientific/model representation; no NASA claim retained without adequate evidence |
| Causal TOD | `Discovering Technology Opportunities with Causal Knowledge` | Check whether mechanism/causal opportunity work predates the project |
| Patent leads | `site:patents.google.com knowledge graph engineering change propagation` | Identify possible claim-overlap leads, not clearance |
| Indian software patents | `site:ipindia.gov.in computer related inventions guidelines 2025 section 3 k` | Use current official rules and distinguish eligibility from novelty |
| Disclosure | `site:wipo.int patent public disclosure before filing novelty` | Decide repository visibility while preserving IP options |
| Venues | Official IEEE TEM, IEEE Access scope/APC/submission pages; AEI scope | Assess topical fit and cost, avoiding invented ranking |
| Pilot grounding | `site:nobelprize.org 2014 physics blue LED white light` | Check the historical enabling example |

## What was opened and read

The source PDF was read in full and rendered. For primary research, the Patent-KG PDF, AM mathematical-KG PDF, and Horber property-impact PDF were accessed and relevant methods/conclusion/appendix portions inspected. Publisher text was available for Liu's hierarchical semantic network study and discovery results for the 2018 semantic change paper. The ASME 2001 abstract was available; its 2004 journal DOI failed to open. Institutional metadata/abstracts supported the Lee and Honores-Marín identity checks.

The official CRI guidelines PDF, WIPO FAQ, W3C PROV-O, IEEE TEM, and IEEE Access pages were accessed. A machine-translated Google Patents record for CN121960672A was opened and its description/date inspected.

Official patent-office legal status and all claim elements were not verified.

## Access limitations

Several ScienceDirect, university, and journal page opens returned 403 or internal errors even where search supplied indexed publisher text. Ren and Zhao's full method, the 2026 R-GCN paper's full text, and the causal-TOD full article remain follow-ups. PTMR-ECM and CN121786237A were discovery leads only. No paywall or access control was bypassed.

Abstract-level records can establish identity and broad method, but cannot support exhaustive negative claims about absent features. Source summaries in this bundle are intentionally limited. Linked full texts remain with their publishers; they are not redistributed in the repository.

## Next gap search before submission

Search the final algorithm terms: applicability-condition composition, constrained inverse design, uncertainty in engineering knowledge graphs, abstention under incomplete physical evidence, and model-based requirements reasoning. Compare the implemented behavior line by line with R07 and R09, then trace their references and newer citations. Expand patent searching around specific claim elements rather than this project name.